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A Hierarchical Spatio-Temporal Refinement Network with residual temporal convolutional networks and closed-form calibration for national-scale short-term load forecasting

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ABSTRACT

Accurate national-scale short-term load forecasting is essential for reliable dispatch, reserve allocation, and cost-efficient operation under weather-driven demand variability. Most existing models either rely mainly on load and calendar inputs or fuse high-dimensional weather grids through simple concatenation, which underuses spatial structure and can increase error at the national scale. This work addresses this gap with the Hierarchical Spatio-Temporal Refinement Network (HST-RefNet), which learns directly from nationwide meteorological fields and historical load without decomposition. The model uses a Spatio-Temporal Representation Temporal Convolutional Network (STR-TCN) to learn dominant spatio-temporal patterns, a Residual-Focused Refinement Temporal Convolutional Network (RFR-TCN) to correct hard regimes, and a lightweight random-feature calibrator with closed-form training to refine remaining nonlinear errors. Iterative training is concentrated in the two TCN stages, while the final calibrator adds low computational overhead through closed-form estimation. The comparative study includes both generic sequence baselines and dedicated spatio-temporal baselines for structured high-dimensional meteorological inputs. On the Pakistan National Transmission and Despatch Company (NTDC) load and National Aeronautics and Space Administration (NASA) Prediction Of Worldwide Energy Resources (POWER) benchmark, HST-RefNet achieves an average Normalized Root Mean Squared Error (NRMSE) of 5.00% and a Coefficient of Determination (R^2) of 0.91 over the first 6 forecast horizons, while reducing NRMSE by 7.09% relative to the strongest spatio-temporal baseline.

1. Introduction

Short-term load forecasting (STLF) supports several United Nations Sustainable Development Goals (SDGs) because it shapes how power systems deliver affordable, reliable, and low-carbon electricity in real time [1]. System operators use STLF to schedule generation and reserves in a manner that advances affordable and clean energy under SDG 7 and strengthens grid infrastructure under SDG 9. Reducing avoidable balancing and redispatch actions from forecast errors also supports climate action under SDG 13 by lowering dependence on carbon-intensive reserve units [2]. Moreover, cost studies report that improvements in forecast accuracy produce measurable operational savings. A one-percent reduction in forecast error for a ten-gigawatt utility yields annual savings of roughly 1.6 million dollars, and forecast errors near five percent already capture most achievable economic gains [3,4]. In

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addition, empirical analyses show that the cost of errors grows when demand approaches peak levels or transmission limits, which concentrates economic losses in a limited number of high-load hours rather than distributing them evenly across the year [2]. These observations motivate forecasting approaches that improve accuracy on average and also in the economically sensitive tail of the error distribution, where many existing methods remain weaker.

Existing STLF research spans several major methodological directions, including classical statistical models, machine learning methods, deep learning architectures, and hybrid or ensemble frameworks. These categories provide a convenient structure for reviewing representative studies in the present work, although the broader STLF literature also includes directions such as transfer learning, meta learning, and Transformer-based forecasting. These directions have also been surveyed from an application-driven, decision-oriented perspective [5]. The statistical branch continues to evolve through models that represent temporal structure more effectively. Authors in [6] develop an online SARIMA model that updates as new observations arrive for short-term electricity-load forecasting. Furthermore, Mado et al. [7] introduce a double-seasonal ARIMA model that captures daily and weekly cycles in half-hourly electricity demand with nested periodicity. In another study, Guo et al. [8] combine ARIMA with Support Vector Regression (SVR), where ARIMA models linear dependence and the SVR component reduces nonlinear residual errors. In addition to this, Shi et al. [9] combine ARIMA with a Random Forest model for sparse load forecasting. On the other hand, Momani et al. [10] compare ARIMA and SARIMA across sampling regimes and show how seasonal structure influences forecasting accuracy across horizons. Together, these contributions trace a progression from single-seasonal models to multi-seasonal and hybrid formulations that use the statistical core as a foundation for data-driven refinement. Despite these developments, the statistical class still relies on linear dynamics and stationarity assumptions and often struggles with nonlinear load responses and abrupt regime changes.

Machine learning methods address these nonlinearities with flexible data-driven mappings. SVR models tuned with meta-heuristic optimizers improve accuracy by adjusting penalty and kernel parameters dynamically. Moreover, Lu and Wang [11] show that an enhanced whale-optimization algorithm that incorporates real-time electricity price achieves consistently lower error than plain SVR and Backpropagation Neural Network baselines. Instance-based methods extend this direction. Hnin et al. [12] propose a K-means and K-nearest-neighbors framework that clusters historical load profiles and assigns new days to these clusters before fitting cluster-specific regressors, including Neural Networks with Bayesian optimization and LSTM with Bayesian optimization. This approach improves error during holiday periods when demand patterns shift quickly. In another line of work, Fan et al. [13] adapt Random Forests with a mean-generating function and a response-surface layer to represent peaks and valleys with limited exogenous input. Furthermore, Jha et al. [14] combine Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Random Forests, and feedforward networks for smart-meter-level forecasting and report improved accuracy across multiple horizons. These machine-learning approaches increase flexibility but still depend on feature engineering and do not fully resolve instability under high-impact regimes.

Deep learning models such as recurrent networks address earlier limitations by learning nonlinear temporal structure directly from raw load and exogenous sequences. Authors in [15] propose a temporal-feature-selection LSTM that incorporates temporal and weather-related features and improves forecasting accuracy. Moreover, authors in [16,17] use recurrent neural-network architectures for short-term load forecasting. Temporal Convolutional Networks (TCNs) extend this direction by using convolutional sequence modeling, and Wang et al. [18] apply a TCN-based framework to industrial-customer load forecasting. These models capture short- and medium-range dependence well, yet their performance often degrades when sequences become long because information is compressed through hidden states and training requires large memory. These constraints motivate architectures that handle long-range correlation, adapt to fast regime changes, and reduce dependence on hand-tuned feature windows.

Transformer and attention-based architectures advance short-term forecasting by learning context-dependent relevance over long sequences, and they can be classified by how they construct tokens from multivariate time-series data. Point-wise models such as the vanilla Transformer and Informer form each token from all variate values at a single time step, so attention operates over temporal tokens that encode multivariate snapshots [19,20]. Patch-wise models such as Patch Time Series Transformer (PatchTST) instead build tokens from short temporal segments of a single variate, which strengthens local temporal modeling and reduces sequence length [21]. Variate-wise models such as iTransformer (ITF) treat each entire univariate series as a token, inverting the axis of attention to focus on cross-variate relations while separate feed-forward layers learn temporal structure [22]. Together, these tokenization schemes broaden the design space of attention mechanisms for time series. However, although attention-based models often outperform recurrent and convolutional networks in many short-term forecasting tasks, they underperform in regimes dominated by rapid fluctuations, such as wind-speed prediction, and they retain substantial computational and memory costs. Related physics-informed spatio-temporal networks already exploit gridded meteorology and physical priors for wind-speed prediction, showing that structured inputs and physics-aware regularization can stabilize training and improve short-horizon accuracy [23]. On the other hand, lightweight Random Vector Functional Link (RVFL) networks train through fixed random projections and closed-form output weights. Authors in [24–26] show that enriched features, deep random layers, and recurrent structures can improve RVFL-based learning and forecasting. These models have been extensively used in recent research for developing various short-term prediction problems as discussed in [27,28].

Recent hybrid models address limitations of single deep learning architectures by combining decomposition, feature learning, and ensemble design. Choubey et al. [29] propose an EMD–transfer learning stacking framework in which Empirical Mode Decomposition separates linear and nonlinear load components, simple base learners such as ARIMA, Prophet, GRU and TFT train on decomposed features, and a linear meta learner integrates their outputs, which improves robustness to nonstationarity and sector specific heterogeneity while it avoids over-parameterization of a single deep network. Moreover, authors in. [30] design a two-stage ICEEMDAN–LSTM scheme for grid-level net load forecasting under high PV penetration in which ICEEMDAN filters high-frequency components with negligible energy and an LSTM forecasts the remaining multiscale modes so model capacity focuses on valley

and peak dynamics rather than high frequency noise. In addition, Han and Zeng [31] address inadequate feature use and weak long-term dependence by coupling a hybrid feature extraction pipeline based on gradient boosted regression trees, Fisher score, and mutual information with a multi-head CNN and a parallel BiLSTM network with lagged parameters, which yields richer feature representations and stronger long-horizon memory without very deep recurrent stacks. In another study, Guo et al. [32] exploit complementarity across learners with a stacking model that integrates ANN, XGBoost, LSTM, stacked LSTM and BiLSTM as base learners and uses Lasso regression as the meta learner while it explicitly includes exogenous factors such as temperature, rainfall and electricity price so a single deep model does not need to learn all multivariate dependencies and nonlinear covariate effects alone. More recently, expert-routing and dynamic-graph designs extend this hybrid direction further. Huang et al. [33] propose a cycle-aware adaptive mixture-of-experts model with a high-demand-sensitive asymmetric loss for electrical load forecasting that routes the forecast across specialized experts and penalizes underestimation more heavily during high-demand periods, while Huang et al. [34] and Zou et al. [35] develop multi-resolution and regime-conditioned horizon-aware mixture-of-experts frameworks for extreme-adaptive and spatio-temporal time-series forecasting. In a related direction, Huang et al. [36] introduce causality-aware multi-graph convolutional networks with critical-node dynamics for electric-vehicle charging-station load forecasting, and Huang et al. [37,38] develop sparse and multi-scale dynamic graph learning for district heat load forecasting, extending an initial sparse-graph design with multi-scale temporal representation. These studies model discrete expert or node-level topologies rather than a continuous nationwide meteorological field. Beyond aggregate load forecasting, Rizi et al. [39] use deep learning-based net-load forecasting that accounts for distributed energy resources and electric vehicles to support power-system flexibility analysis.

Recent work also emphasizes preprocessing quality, adaptive learning, and weather-aware hybrid design in short-term load forecasting. Iqbal et al. [40] propose a hybrid deep learning framework that combines data standardization and cleansing with bidirectional recurrent modeling, showing that careful preprocessing can materially improve forecasting accuracy. In a subsequent study, Iqbal et al. [41] introduce a self-adaptive learning framework for residential and commercial STLF under stochastic and weather-sensitive demand conditions. At the building scale, Khan et al. [42] combine preprocessing, ConvLSTM, bidirectional GRU, and multilayer perceptron layers to forecast both power consumption and generation while incorporating weather information. These studies strengthen the case for adaptive hybrid and weather-aware forecasting. However, they still focus on local or building-scale settings and do not learn directly from nationwide structured meteorological grids.

Despite these advances, many short-term load forecasting studies still rely on low-dimensional inputs and operate at limited spatial scales. Authors in [6] use online Seasonal Autoregressive Integrated Moving Average (SARIMA) with historical load dynamics, while Mado et al. [7] extend this direction with a double-seasonal ARIMA formulation for daily and weekly cycles. Guo et al. [8] further combine ARIMA with Support Vector Regression (SVR), yet the input design still remains fundamentally low-dimensional. In deep learning, authors in [15] improve load prediction through temporal feature selection in Long Short-Term Memory (LSTM), and authors in [18] use Temporal Convolutional Networks (TCNs) for efficient sequence modeling. Transformer-based variants such as Informer [20], PatchTST [21], and ITF [22] also focus primarily on multivariate temporal tokens rather than explicitly structured national meteorological grids. When meteorological variables are included, they are often represented as aggregated factors at one or a few locations, as in the stacking model of Guo et al. [32], the residential forecasting study of Cao et al. [43], and the building-level analysis of Koukaras et al. [44]. By contrast, related spatio-temporal studies in adjacent forecasting problems show that structured gridded meteorology can be valuable; for example, Aslam et al. [23] use physics-informed spatio-temporal learning with gridded weather fields for short-term wind prediction. Moreover, efficient random-feature and shallow hybrid designs remain attractive when computational cost matters [27,28]. Decomposition-based hybrids such as the EMD-transfer learning stacking framework of Choubey et al. [29] and the ICEEMDAN-LSTM model in [30] improve robustness, but they introduce additional preprocessing and reconstruction choices. As a result, existing approaches still rarely learn directly from nationwide gridded meteorological fields while also modeling residual errors through an explicit hierarchical refinement mechanism.

To address these challenges, this work proposes the Hierarchical Spatio-Temporal Refinement Network (HST-RefNet), a forecasting architecture that learns from nationwide meteorological fields and load histories without dimensionality reduction or decomposition. The model ingests Pakistan-wide meteorological grids at 100 km spatial resolution together with historical load and first processes them through a Spatio-Temporal Representation TCN (STR-TCN), which captures the dominant temporal structure while preserving information from all grid cells. A second module, the Residual-Focused Refinement TCN (RFR-TCN), learns to predict the residual errors of STR-TCN so that it allocates capacity to horizons and operating regimes that the first stage does not capture accurately. Unlike decomposition-based hybrids that split the signal before forecasting, HST-RefNet performs post-hoc hierarchical refinement on learned residuals, allowing each stage to concentrate on errors that earlier stages systematically miss. The final stage employs an Embedded Deep Random Feature Calibrator (edRFC), a lightweight edRVFL-based correction layer that integrates the forecasts of the two TCN modules and computes a closed-form nonlinear adjustment. Together, these components define a unified forecasting framework for national-scale STLF that combines direct learning from nationwide gridded meteorological fields, hierarchical refinement through two temporally specialized TCN stages, and a lightweight closed-form calibration layer that consolidates the outputs of both stages. This design improves the representation of large-scale weather-driven demand patterns, strengthens correction of hard residual regimes, and retains practical computational efficiency for short-horizon forecasting. The contributions of this work are summarized as follows:

- This study develops HST-RefNet, a national-scale short-term load forecasting architecture that learns jointly from high-dimensional gridded meteorological fields and historical system load.
- It introduces a hierarchical two-stage temporal refinement strategy in which STR-TCN captures the dominant spatio-temporal load–weather mapping and RFR-TCN specializes in the residual regimes that remain difficult after the first-stage forecast.

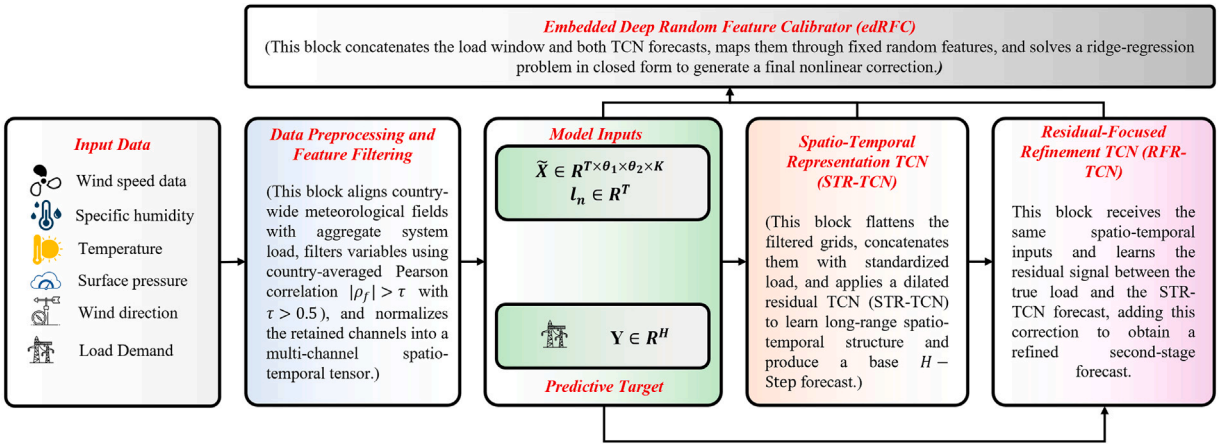


Fig. 1. Overall architecture of the proposed HST-RefNet framework, illustrating the data flow from meteorological grids and historical load through STR-TCN, RFR-TCN, and the edRFC calibration stage.

- It integrates an embedded deep random-feature calibrator that combines the historical load window with the outputs of both TCN stages to provide an efficient closed-form nonlinear correction.

The rest of the paper is organized as follows: Section 2 presents the proposed methodology, Section 3 reports the experimental results and comparative evaluation, and Section 4 offers concluding remarks.

2. Proposed methodology

This section presents the proposed Hierarchical Spatio-Temporal Refinement Network (HST-RefNet), a three-stage framework designed to model nationwide meteorological fields together with historical load and to refine predictions across increasingly specialized modules. The first stage constructs a structured spatio-temporal representation by filtering and normalizing high-dimensional meteorological grids and aligning them with historical load windows. The second stage, STR-TCN, learns the dominant temporal structure by integrating grid features with past load through dilated causal convolutions. The third stage, RFR-TCN, focuses on hard-to-model cases by learning the residual errors of the first-stage predictor. Finally, a lightweight Embedded Deep Random Feature Calibrator (edRFC) produces a closed-form nonlinear adjustment that corrects remaining mismatches. The overall architecture is summarized in the block diagram shown in Fig. 1.

2.1. Data preprocessing and feature filtering

The first step of the proposed methodology constructs a structured spatio-temporal representation of the meteorological environment and filters out redundant variables before model training. This stage transforms raw meteorological fields into a normalized multi-channel tensor by discretizing the spatial domain, evaluating feature relevance using Pearson correlation, retaining only relevant variables, and scaling the retained features over time and space. The spatial domain of interest is denoted by $\Omega \subset \mathbb{R}^2$ and is discretized into a uniform grid of spatial resolution S, producing θ_1 horizontal and θ_2 vertical grid points:

$$\mathcal{G} = \{(x_i, y_j) \mid i = 1, \ldots, \theta_1,\ j = 1, \ldots, \theta_2\}, \tag{1}$$

where $\mathcal{G}$ is the set of sampling points and $|\mathcal{G}| = \theta_1\theta_2$ is the total number of grid locations. At each spatial location (x_i, y_j) and time t, we collect F meteorological variables forming the vector

$$\mathbf{v}_t(i,j) = [v_t^{(1)}(i,j),\ v_t^{(2)}(i,j),\ \ldots,\ v_t^{(F)}(i,j)], \tag{2}$$

where $v_t^{(f)}(i,j)$ denotes the value of feature f at time t and grid location (i,j), including temperature, humidity, wind speed, wind direction, and air pressure. To quantify feature relevance, we compute a grid-cell-wise Pearson correlation coefficient between each variable at (i,j) and the target load sequence y_t:

$$r_f(i,j) = \frac{\sum_t (v_t^{(f)}(i,j) - \bar{v}^{(f)}(i,j))(y_t - \bar{y})}{\sqrt{\sum_t (v_t^{(f)}(i,j) - \bar{v}^{(f)}(i,j))^2}\ \sqrt{\sum_t (y_t - \bar{y})^2}}, \tag{3}$$

where $r_f(i,j)$ is the correlation for feature f at grid cell (i,j), $\bar{v}^{(f)}(i,j)$ is the mean of that feature at (i,j), and $\bar{y}$ is the mean of the load series. We then summarize the national relevance of each meteorological variable by averaging its correlation field over grid

cells that fall inside the country boundary:

$$\rho_f = \frac{1}{|\mathcal{G}_{\text{ctry}}|} \sum_{(i,j)\in\mathcal{G}_{\text{ctry}}} r_f(i,j), \tag{4}$$

where $\mathcal{G}_{\text{ctry}} \subset \mathcal{G}$ denotes the set of grid points located within the country boundary and $|\mathcal{G}_{\text{ctry}}|$ is its cardinality. A feature is retained if the absolute value of this country-averaged correlation exceeds a threshold τ:

$$\mathcal{F}_{\text{sig}} = \{f \mid |\rho_f| > \tau\}, \qquad K = |\mathcal{F}_{\text{sig}}| < F. \tag{5}$$

where $\mathcal{F}_{\text{sig}}$ is the set of significant features and K is the number of selected variables. The Pearson-based screening is used as a conservative dimensionality-control step rather than as a complete characterization of meteorology–load dependence. Because the prediction target is country-level aggregate load and each retained meteorological variable contributes a full national grid channel, the threshold $\tau = 0.5$ is used to retain only variables with moderate-to-strong spatially persistent national association, while the subsequent STR-TCN and RFR-TCN learn nonlinear and lagged dependencies from the retained spatio-temporal inputs. For each retained feature $f \in \mathcal{F}_{\text{sig}}$, a spatio-temporal tensor is formed across a prediction window of length T:

$$\mathbf{X}^{(f)} = [v_t^{(f)}(i,j)]_{t=1,\dots,T;\ i=1,\dots,\theta_1;\ j=1,\dots,\theta_2} \in \mathbb{R}^{T\times\theta_1\times\theta_2}, \tag{6}$$

where $\mathbf{X}^{(f)}$ contains the temporal evolution of feature f over all spatial grid points. The full input tensor is constructed by concatenating all retained features along the feature channel:

$$\mathbf{X} = \text{concat}(\mathbf{X}^{(f_1)}, \dots, \mathbf{X}^{(f_K)}) \in \mathbb{R}^{T\times\theta_1\times\theta_2\times K}. \tag{7}$$

where $\mathbf{X}$ represents the complete spatio-temporal input field. Finally, each retained feature is normalized using min–max scaling computed on the training set:

$$\tilde{v}_t^{(f)}(i,j) = \frac{v_t^{(f)}(i,j) - v_{\min}^{(f)}}{v_{\max}^{(f)} - v_{\min}^{(f)}}, \tag{8}$$

where $v_{\min}^{(f)}$ and $v_{\max}^{(f)}$ denote the minimum and maximum values of feature f in the training period. The resulting normalized model input is

$$\tilde{\mathbf{X}} \in \mathbb{R}^{T\times\theta_1\times\theta_2\times K}. \tag{9}$$

In parallel with the meteorological fields, we consider the aggregate load time series $\{y_t\}_{t=1}^{T_{\text{tot}}}$ over the same period. For each time index t, the load is standardized using training-set statistics,

$$\tilde{y}_t = \frac{y_t - \mu_{\text{load}}}{\sigma_{\text{load}}}, \tag{10}$$

where μ_{load} and σ_{load} are the mean and standard deviation of the load over the training set. For each training example indexed by n, we form an input load window of length T and an H-step forecast target for the whole system,

$$\mathscr{E}_n = [\tilde{y}_{t_n-T+1}, \dots, \tilde{y}_{t_n}]^\top \in \mathbb{R}^T, \qquad \mathbf{y}_n^{(H)} = [y_{t_n+1}, \dots, y_{t_n+H}]^\top \in \mathbb{R}^H, \tag{11}$$

where $\mathscr{E}_n$ is the standardized historical load window for sample n and $\mathbf{y}_n^{(H)}$ is the corresponding H-hour ahead load vector.

2.2. Spatio-Temporal Representation TCN (STR-TCN)

The second step of the proposed methodology encodes the normalized spatio-temporal input into a compact representation that captures long-range temporal dependencies under heterogeneous spatial and socio-economic conditions. In a country-scale system, some regions such as metropolitan areas contribute disproportionately to the aggregate load, while rural and small-town regions exhibit lower but still seasonally varying demand. Moreover, fuel mix and end-use patterns differ across locations; for example, areas with extensive natural-gas infrastructure may use less electricity for winter heating than regions that depend primarily on electric heating, and income-driven spending habits further modulate demand response to weather. Temperature and other meteorological variables also evolve non-uniformly across the domain, with some zones operating below freezing while others remain near 30°C during the same period. A temporal model must therefore integrate slowly varying large-scale gradients with sharp local regime changes while remaining stable and efficient on long sequences. Temporal Convolutional Networks (TCNs) are well suited to this setting because they use causal convolutions with dilation to build a controllable temporal receptive field, support parallel computation over time, and maintain stable gradients through residual connections.

Given the normalized input tensor $\tilde{\mathbf{X}} \in \mathbb{R}^{T\times\theta_1\times\theta_2\times K}$ in (9) and the standardized load sequence $\tilde{y}_t$ in (10), we construct for each time step t a joint feature vector

$$\mathbf{u}_t = \left[\text{vec}\left(\tilde{\mathbf{X}}_{t,:,:,:}\right)^\top,\ \tilde{y}_t\right]^\top \in \mathbb{R}^{C+1}, \tag{12}$$

where $\text{vec}(\cdot)$ stacks all spatial locations and feature channels into a single vector and $C = \theta_1\theta_2 K$ is the total number of grid-based channels. Stacking all temporal steps yields the input sequence

$$\mathbf{U} = \left[\mathbf{u}_1^\top, \mathbf{u}_2^\top, \dots, \mathbf{u}_T^\top\right]^\top \in \mathbb{R}^{T\times(C+1)}. \tag{13}$$

A TCN layer applies a one-dimensional dilated causal convolution along the temporal axis. Let $\mathbf{h}_t^{(0)} = \mathbf{u}_t$ denote the input at time t. For layer ℓ, the dilated causal convolution with kernel size K_c and dilation factor d_ℓ is defined as

$$\left(\mathbf{W}^{(\ell)} *_{d_\ell} \mathbf{h}^{(\ell-1)}\right)_t = \sum_{k=0}^{K_c-1} \mathbf{W}_k^{(\ell)} \mathbf{h}_{t-d_\ell k}^{(\ell-1)}, \tag{14}$$

where $\mathbf{W}_k^{(\ell)} \in \mathbb{R}^{C_\ell \times C_{\ell-1}}$ denotes the kth convolution kernel at layer ℓ, $C_{\ell-1}$ and C_ℓ are the input and output channel dimensions, and the summation respects causality by using only past or current time steps.

Each TCN layer then computes its hidden representation with a nonlinear activation and bias,

$$\tilde{\mathbf{h}}_t^{(\ell)} = \phi\Big(\left(\mathbf{W}^{(\ell)} *_{d_\ell} \mathbf{h}^{(\ell-1)}\right)_t + \mathbf{b}^{(\ell)}\Big), \tag{15}$$

where $\phi(\cdot)$ is a pointwise nonlinearity such as ReLU and $\mathbf{b}^{(\ell)} \in \mathbb{R}^{C_\ell}$ is the bias vector of layer ℓ.

To improve optimization stability and gradient flow on long sequences, STR-TCN adopts residual connections. The residual update at layer ℓ is given by

$$\mathbf{h}_t^{(\ell)} = \mathbf{h}_t^{(\ell-1)} + \tilde{\mathbf{h}}_t^{(\ell)}, \tag{16}$$

where $\mathbf{h}_t^{(\ell)} \in \mathbb{R}^{C_\ell}$ is the output of layer ℓ at time t. If $C_{\ell-1} \neq C_\ell$, a linear projection is applied to $\mathbf{h}_t^{(\ell-1)}$ so that the residual sum is well defined.

Stacking L such dilated residual layers yields a temporal receptive field that grows with depth and dilation. For exponentially increasing dilations $d_\ell = 2^{\ell-1}$, the effective receptive field at the final layer is

$$R_L = 1 + (K_c - 1)\left(2^L - 1\right), \tag{17}$$

where R_L is the number of past time steps that influence the output at a given time index. This design allows STR-TCN to capture both short-range dynamics, such as hour-to-hour fluctuations driven by local weather and usage patterns, and long-range dependencies, such as gradual seasonal shifts and persistent socio-economic behaviors, without resorting to very deep recurrent stacks.

The STR-TCN output at time t is obtained from the final hidden state by a linear readout,

$$\mathbf{z}_t = \mathbf{W}_o \mathbf{h}_t^{(L)} + \mathbf{b}_o, \tag{18}$$

where $\mathbf{W}_o \in \mathbb{R}^{C_z \times C_L}$ and $\mathbf{b}_o \in \mathbb{R}^{C_z}$ are the output projection weights and bias, and C_z is the dimension of the STR-TCN representation. This representation aggregates information from all spatial locations and significant meteorological features over a large temporal window, enabling downstream refinement modules to distinguish the contribution of high-impact regions, seasonal regimes, and socio-economic patterns to the aggregate load.

2.3. Residual-Focused Refinement TCN (RFR-TCN)

While STR-TCN provides a strong spatio-temporal representation, forecasting over a large and heterogeneous domain inevitably produces examples that are more difficult to model than others. National-scale load patterns are shaped by multiple interacting factors including demographic density, income levels, industrial concentration, access to alternative heating fuels, and highly uneven meteorological conditions. These variables do not evolve uniformly. Metropolitan regions often exhibit sharp ramps driven by commercial and residential activity, whereas rural regions follow smoother patterns; similarly, temperature may range from sub-zero in one zone to nearly 30°C in another during the same period. As a result, a subset of time intervals contains abrupt regime transitions, atypical feature combinations, or rare demand conditions that STR-TCN may not capture with equal fidelity. These hard examples disproportionately influence forecasting risk because they frequently correspond to peak or near-peak operating hours.

To better accommodate these challenging regimes, the proposed framework introduces a second-stage refinement network, RFR-TCN, which learns to predict the residual error of STR-TCN rather than the load itself. Let $\hat{y}_t^{\mathrm{STR}}$ denote the first-stage prediction of STR-TCN at time t. The residual error is defined as

$$r_t = y_t - \hat{y}_t^{\mathrm{STR}}, \tag{19}$$

where y_t is the ground-truth load. The residual sequence $\{r_t\}$ captures precisely the discrepancies that STR-TCN fails to model, including abrupt demand surges, uncommon weather-load interactions, and localized anomalies arising from socio-economic conditions, holiday effects, or unexpected behavioral shifts.

RFR-TCN takes as input the same spatio-temporal tensor used by STR-TCN and predicts the corresponding residual component. Let $\mathbf{U}$ be the temporal sequence defined in (13). A second dilated convolutional stack processes this sequence, giving the refinement representation

$$\mathbf{h}_{\mathrm{RFR},t}^{(\ell)} = \mathbf{h}_{\mathrm{RFR},t}^{(\ell-1)} + \phi\Big(\left(\mathbf{W}_{\mathrm{RFR}}^{(\ell)} *_{d_\ell} \mathbf{h}_{\mathrm{RFR}}^{(\ell-1)}\right)_t + \mathbf{b}_{\mathrm{RFR}}^{(\ell)}\Big), \tag{20}$$

where $\mathbf{W}_{\mathrm{RFR}}^{(\ell)}$ and $\mathbf{b}_{\mathrm{RFR}}^{(\ell)}$ are the convolutional weights and biases of the refinement network, and the dilated convolution $*_{d_\ell}$ follows the definition in (14). The predicted residual at time t is obtained through a linear projection:

$$\hat{r}_t = \mathbf{w}_{\mathrm{RFR}}^{\top} \mathbf{h}_{\mathrm{RFR},t}^{(L)} + b_{\mathrm{RFR}}, \tag{21}$$

where $\mathbf{w}_{\mathrm{RFR}}$ and b_{RFR} are trainable output parameters. The refinement prediction $\hat{r}_t$ is combined with the STR-TCN estimate to produce the stage-two forecast:

$$\hat{y}_t^{(2)} = \hat{y}_t^{\mathrm{STR}} + \hat{r}_t. \tag{22}$$

This two-stage design offers two major advantages. First, by specializing on residual structure, RFR-TCN concentrates model capacity on hard examples that STR-TCN underfits, including rare temporal regimes and spatially localized anomalies. Second, residual learning stabilizes training by reducing the dynamic range of the prediction task; the residuals are generally smaller in magnitude and often reflect simpler high-frequency corrections rather than full-scale load dynamics. Together, these properties enable RFR-TCN to improve accuracy particularly in high-impact periods where forecasting errors carry disproportionate economic and operational consequences.

2.4. Embedded Deep Random Feature Calibrator (edRFC)

Even after the two-stage refinement provided by STR-TCN and RFR-TCN, certain forecasting errors may persist due to highly localized nonlinear effects, rare meteorological–load interactions, or abrupt demand fluctuations that occur infrequently in the training data. These residual patterns often arise from complex socio-economic behavior such as sudden commercial surges, weekend–weekday transitions, or abnormal weather anomalies that influence only a subset of regions. Deep temporal models may underrepresent such sparse patterns because their learning dynamics prioritize globally consistent structure. To correct these remaining discrepancies without introducing additional deep or computationally intensive modules, the proposed framework incorporates an Embedded Deep Random Feature Calibrator (edRFC) as a final lightweight nonlinear adjustment layer.

For each training example n, we construct a calibration feature vector by concatenating the standardized historical load window $\boldsymbol{\ell}_n$ from (11) with the H-step forecasts produced by the first and second TCN stages, denoted by $\hat{\mathbf{y}}_n^{(1)}$ and $\hat{\mathbf{y}}_n^{(2)}$ respectively:

$$\mathbf{s}_n = \left[\boldsymbol{\ell}_n^{\top},\ \hat{\mathbf{y}}_n^{(1)\top},\ \hat{\mathbf{y}}_n^{(2)\top},\ 1\right]^{\top} \in \mathbb{R}^{T+2H+1}, \tag{23}$$

where the final component is an explicit bias term. This construction aligns with the implementation that feeds the concatenation of the load window and both TCN outputs into the edRFC module.

Random Vector Functional Link (RVFL) networks generate nonlinear transformations using fixed random projections, eliminating the need to backpropagate through these layers. The random feature transformation applies a fixed nonlinear expansion,

$$\mathbf{h}_n = \phi\left(\mathbf{W}_{\mathrm{rf}}\,\mathbf{s}_n\right), \tag{24}$$

where $\mathbf{W}_{\mathrm{rf}} \in \mathbb{R}^{H_{\mathrm{rf}} \times (T+2H+1)}$ is a randomly initialized and fixed weight matrix, H_{rf} is the number of random neurons, and $\phi(\cdot)$ is a nonlinear activation such as ReLU or sigmoid. This yields a random feature embedding $\mathbf{h}_n \in \mathbb{R}^{H_{\mathrm{rf}}}$ for each example.

Stacking the embeddings over all N training samples gives

$$\mathbf{H} = \left[\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N\right]^{\top} \in \mathbb{R}^{N \times H_{\mathrm{rf}}}, \tag{25}$$

where each row corresponds to the random feature representation of one training example. The output weights of the calibrator minimize the ridge-regularized least-squares objective,

$$\boldsymbol{\beta}^{\star} = \arg\min_{\boldsymbol{\beta}} \left(\|\mathbf{H}\boldsymbol{\beta} - \mathbf{y}^{(H)}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right), \tag{26}$$

where $\lambda > 0$ is the regularization hyperparameter and $\mathbf{y}^{(H)} = [\mathbf{y}_1^{(H)\top}, \ldots, \mathbf{y}_N^{(H)\top}]^{\top}$ stacks all multi-step targets. The optimal weights admit the closed-form solution

$$\boldsymbol{\beta}^{\star} = \left(\mathbf{H}^{\top}\mathbf{H} + \lambda \mathbf{I}\right)^{-1} \mathbf{H}^{\top}\mathbf{y}^{(H)}, \tag{27}$$

where $\mathbf{I}$ is the identity matrix of appropriate dimension. For a new example n, the calibrated H-step forecast is

$$\hat{\mathbf{y}}_n^{\mathrm{final}} = \hat{\mathbf{y}}_n^{(2)} + \mathbf{h}_n^{\top}\boldsymbol{\beta}^{\star}, \tag{28}$$

where the additive correction $\mathbf{h}_n^{\top}\boldsymbol{\beta}^{\star}$ captures remaining nonlinear discrepancies that STR-TCN and RFR-TCN do not fully resolve. This calibration layer introduces expressive nonlinear correction at negligible computational cost, guarantees a convex optimization step through its closed-form training procedure, and targets precisely the residual patterns that deep temporal networks struggle to represent reliably. The pseudo-code of the proposed HST-RefNet framework is provided in Algorithm 1.

3. Results and comparison

This section reports results and comparisons for HST-RefNet on national-scale STLF using Pakistan NTDC load aligned with NASA POWER meteorological grids. The dataset description summarizes spatial coverage, hourly sampling, forecast horizon, and chronological splits to ground the evaluation in an operational setting. The experimental setup keeps the optimization budget consistent across baselines and assesses accuracy with NRMSE, NMAE, MAPE, WAPE, SMAPE, and R2. An ablation of the residual TCN stages and the edRVFL calibrator, together with horizon-wise analyses and Wilcoxon tests, attributes the observed gains to specific components.

Algorithm 1: HST-RefNet: Hierarchical Spatio-Temporal Refinement Network

Input: Historical load: $\{y_t\}$, Meteorological grids: $\{v_t^{(f)}(i,j)\}$
Window length: T, Forecast horizon: H
Correlation threshold: τ, TCN layers: L, Kernel size: K_c
Random feature dimension: H_{rf}, Regularization: λ
Output: Final forecasts: $\hat{y}_n^{\text{final}}$ for $n = 1, \ldots, N_{\text{test}}$

Stage 1: Data Preprocessing ; // Normalize and select features
Normalize load series: $\tilde{y}_t = (y_t - \mu_{\text{load}})/\sigma_{\text{load}}$;
Compute correlation ρ_f for each meteorological feature f ;
Select features: $\mathcal{F}_{\text{sig}} = \{f \mid |\rho_f| > \tau\}$;
Construct normalized input tensor $\tilde{\mathbf{X}}$ and load windows $\mathscr{E}_n$;
Stage 2: STR-TCN Training ; // Learn dominant patterns
Initialize STR-TCN with random weights ;
while *not converged* **do**
  $\hat{\mathbf{y}}^{\text{STR}} \leftarrow \text{STR-TCN}(\tilde{\mathbf{X}}, \mathscr{E}_n)$; // Forward pass
  $\mathcal{L}_{\text{STR}} \leftarrow \text{MSE}(\mathbf{y}, \hat{\mathbf{y}}^{\text{STR}})$; // Compute loss
  Update STR-TCN parameters via backpropagation ;
end
Stage 3: RFR-TCN Training ; // Refine residuals
$\mathbf{r} \leftarrow \mathbf{y} - \hat{\mathbf{y}}^{\text{STR}}$; // Compute residuals
Initialize RFR-TCN with random weights ;
while *not converged* **do**
  $\hat{\mathbf{r}} \leftarrow \text{RFR-TCN}(\tilde{\mathbf{X}}, \mathscr{E}_n)$; // Predict residuals
  $\mathcal{L}_{\text{RFR}} \leftarrow \text{MSE}(\mathbf{r}, \hat{\mathbf{r}})$; // Residual loss
  Update RFR-TCN parameters via backpropagation ;
end
$\hat{\mathbf{y}}^{(2)} \leftarrow \hat{\mathbf{y}}^{\text{STR}} + \hat{\mathbf{r}}$; // Stage 2 forecast
Stage 4: edRFC Training ; // Closed-form calibration
Construct feature vector: $\mathbf{s}_n = [\mathscr{E}_n, \hat{y}_n^{\text{STR}}, \hat{y}_n^{(2)}, 1]$;
Generate random features: $\mathbf{h}_n = \phi(\mathbf{W}_{\text{rf}}\mathbf{s}_n)$;
$\mathbf{H} \leftarrow [\mathbf{h}_1, \ldots, \mathbf{h}_{N_{\text{train}}}]^\top$;
$\boldsymbol{\beta}^\star \leftarrow (\mathbf{H}^\top\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}^\top\mathbf{y}$; // Closed-form solution
Stage 5: Forecasting ; // Generate final predictions
for *each test sample* n **do**
  $\hat{\mathbf{y}}^{\text{STR}} \leftarrow \text{STR-TCN}(\tilde{\mathbf{X}}_n, \mathscr{E}_n)$;
  $\hat{\mathbf{r}} \leftarrow \text{RFR-TCN}(\tilde{\mathbf{X}}_n, \mathscr{E}_n)$;
  $\hat{y}^{(2)} \leftarrow \hat{\mathbf{y}}^{\text{STR}} + \hat{\mathbf{r}}$;
  $\mathbf{s}_n \leftarrow [\mathscr{E}_n, \hat{\mathbf{y}}^{\text{STR}}, \hat{y}^{(2)}, 1]$;
  $\mathbf{h}_n \leftarrow \phi(\mathbf{W}_{\text{rf}}\mathbf{s}_n)$;
  $\hat{y}_n^{\text{final}} \leftarrow \hat{y}^{(2)} + \mathbf{h}_n^\top\boldsymbol{\beta}^\star$;
end

3.1. Dataset description

To evaluate the performance of the proposed HST-RefNet, this work uses a national-scale STLF dataset that combines country-level system demand with gridded meteorological fields. The study obtains aggregate hourly load from the National Transmission and Despatch Company (NTDC) of Pakistan. NTDC operates the high-voltage transmission network and coordinates bulk power dispatch across most regions of the country. This setting captures nationwide demand formation rather than localized feeder effects. Pakistan offers a useful testbed because large regional differences in climate, urban activity, and industrial structure create spatially heterogeneous drivers that low-dimensional load-and-calendar inputs do not represent well at national scale.

The study retrieves reanalysis predictors from the National Aeronautics and Space Administration (NASA) Prediction Of Worldwide Energy Resources (POWER) archive based on the Modern-Era Retrospective analysis for Research and Applications version 2 (MERRA-2). The extracted fields include wind speed at 50 m (WS50M) in m/s, wind direction at 50 m (WD50M) in degrees, surface pressure (PS) in kPa, specific humidity at 2 m (QV2M) in g/kg, and air temperature at 2 m (T2M) in °C. The POWER records provide hourly values in local standard time (LST) and the native spatial support corresponds to an approximately 0.5×0.625 degree latitude–longitude region, which aligns with the gridded national tensor used in this study. The study constructs a latitude–longitude grid over Pakistan aligned with the 0.50° POWER lattice with $\text{LAT}_{\text{min}} = 23.69$, $\text{LAT}_{\text{max}} = 37.13$, $\text{LON}_{\text{min}} = 60.87$, and $\text{LON}_{\text{max}} = 77.84$. After snapping bounds to the 0.50° grid, the domain covers 29 latitude points and 36 longitude points for 1044 grid cells. The study aligns POWER features and NTDC load by identical hourly timestamps, then forms model inputs by pairing each load window with the corresponding country-wide meteorological tensor at the same time indices.

The aligned record spans January 1, 2015, at 00:00 to May 1, 2020, at 23:00 and yields $N = 46{,}752$ hourly samples. The study applies a chronological split with no overlap. The first 80% hours form the training set, the next 10% hours form the validation set, and the last 10% hours form the test set. This design preserves temporal causality and limits information leakage from later periods. For supervised forecasting, each example uses a 48-hour meteorological sequence together with a 48-hour historical load window to predict the next 6 h. Accordingly, the effective number of chronological supervised samples in each split is computed as $N_{\text{split}} - T - H + 1$, which yields 37,348 training samples, 4622 validation samples, and 4623 test samples. No input-target window is allowed to cross split boundaries (see Table 1).

Fig. 2 summarizes seasonal load behavior using bars for monthly mean load and an orange variance curve computed from hourly observations across 2015 to 2020. The mean load reaches its minimum in February at 9.54 GW and its maximum in July at 16.80 GW, which is a 75.99% increase. The variance peaks in September at 6.71 GW^2 and drops to its lowest level in November at 1.45 GW^2. This seasonal amplitude and month-dependent uncertainty support a rigorous evaluation of hierarchical spatio-temporal refinement under country-wide meteorological forcing.

Table 1
Chronological split over the full record of $N = 46,752$ h and effective supervised samples after applying $T = 48$ and $H = 6$ with no cross-split windows.

Split	Share	Hours	Supervised samples
Training	80%	37,401	37,348
Validation	10%	4675	4622
Test	10%	4676	4623
Total	100%	46,752	46,593

Table 2
Country-averaged Pearson correlation between hourly system load and each meteorological predictor over the full record.

	T2M (°C)	QV2M (g/kg)	PS (kPa)	WD50M (deg)	WS50M (m/s)
Mean Pearson r	0.73	0.55	−0.61	0.09	0.07

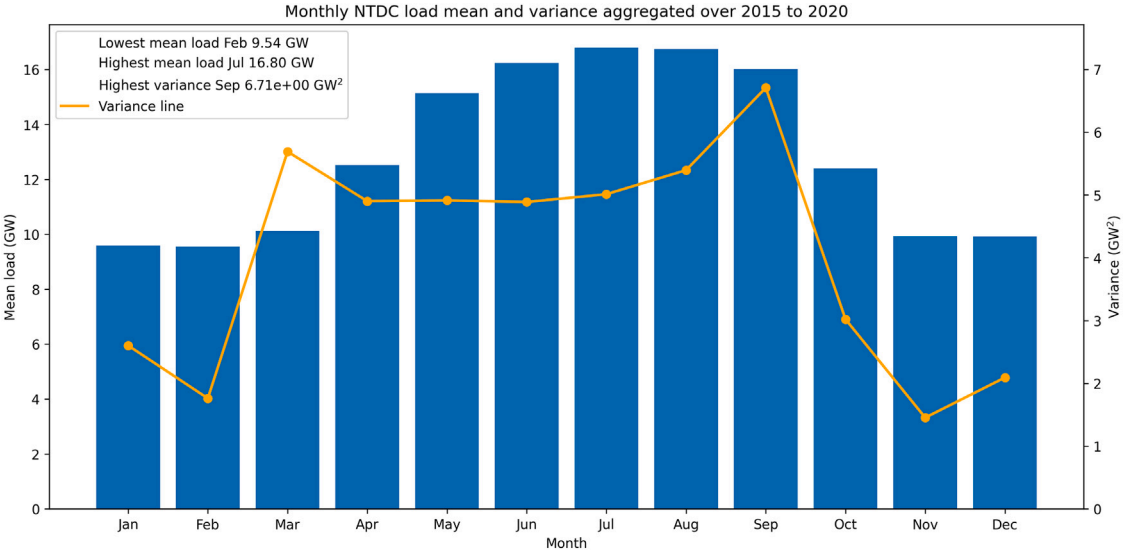


Fig. 2. Monthly mean system load and variance aggregated over 2015–2020. Bars show mean load in GW and the orange line shows variance in GW^2.

3.2. Meteorology–load correlation and feature selection

Before training HST-RefNet, the study examines how each meteorological predictor co-varies with country-level system demand. Using the grid-cell-wise Pearson coefficient in (3) and the country-averaged statistic in (4), the study computes correlation fields between hourly load y_t and each variable (WS50M, WD50M, PS, QV2M, T2M) over the full aligned record. The fields are then averaged over all grid cells inside the national boundary to obtain a single summary coefficient ρ_f for each predictor. Table 2 reports these country-averaged correlations.

Air temperature at 2 m (T2M) exhibits the strongest positive association with load, followed by specific humidity at 2 m (QV2M). Surface pressure (PS) shows a strong negative association, whereas wind speed (WS50M) and wind direction (WD50M) have much weaker correlations. In subsequent experiments, the feature-selection threshold in (5) is set to $\tau = 0.5$, so variables with $|\rho_f| \geq \tau$ (T2M, QV2M, and PS) are retained as meteorological channels and WS50M and WD50M are discarded.

Fig. 3 visualizes the correlation fields for the three retained variables. T2M and QV2M show broadly positive correlations across most of the domain, while PS exhibits a pronounced negative band in regions that contribute heavily to national demand. These spatial patterns support the use of temperature, humidity, and pressure as the primary meteorological drivers in HST-RefNet.

3.3. Experimental setup

The study integrates aligned NTDC load and POWER meteorological variables into a unified spatio-temporal input and evaluates HST-RefNet and all baselines under identical chronological train–validation–test partitions. Each model uses the previous $T = 48$ h to predict the next $H = 6$ h, targeting operational short-horizon forecasting while retaining sensitivity to rapid weather-driven demand variations. Hyperparameters are optimized with the Tree-structured Parzen Estimator (TPE) over 100 trials per model, except

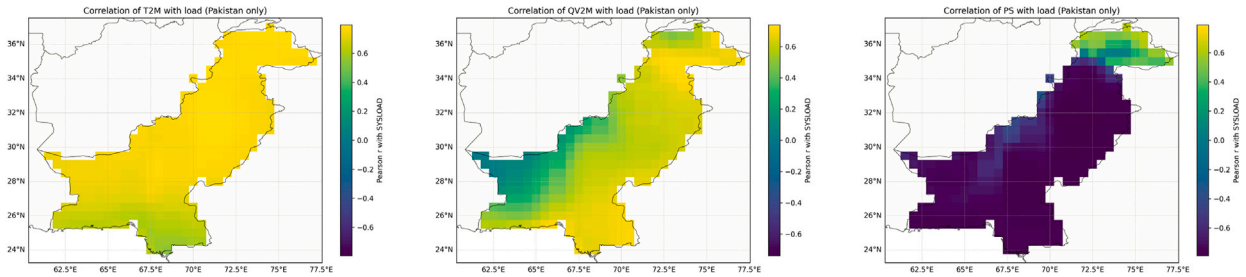


Fig. 3. Grid-cell-wise Pearson correlation between hourly system load and the three retained meteorological variables (T2M, QV2M, PS).

Table 3
Experimental setup used for all models unless a method specifies a dedicated optimizer.

Item	Setting
Input window length	$T = 48$ h
Forecast horizon	$H = 6$ h
Hyperparameter search	TPE with 100 trials per model
Selection criterion	Validation NRMSE averaged over first 6 horizons
Epoch budget	2000
Loss	MSE
Batch size	16
Initial learning rate	1×10^{-4}
Learning rate control	Adaptive adjustment
Early stopping	Patience 10
Hardware	NVIDIA RTX 3080 Ti

methods with dedicated optimization schemes, which retain their original procedures. Each candidate is trained on the training split and ranked by the lowest validation NRMSE averaged across the six forecast horizons. The search spans architecture-specific parameters, including hidden width, layer depth, kernel size or attention dimension, dropout, learning rate, activation function, random-feature dimensions, and ridge regularization, while the common training protocol uses Mean Squared Error (MSE) loss, batch size 16, an initial learning rate of 1×10^{-4}, adaptive learning-rate adjustment, and up to 2000 epochs with early stopping of patience 10. The selected configuration is subsequently retrained and evaluated exclusively on the held-out test set. All experiments run on an NVIDIA RTX 3080 Ti, with Tables 3 and 4 summarizing the common training settings and hyperparameter search spaces, respectively.

3.4. Comparison metrics

To assess forecasting accuracy from complementary error perspectives, this study evaluates all models using six standard metrics: Normalized Root Mean Squared Error (NRMSE), Normalized Mean Absolute Error (NMAE), Mean Absolute Percentage Error (MAPE), Weighted Absolute Percentage Error (WAPE), Symmetric Mean Absolute Percentage Error (SMAPE), and the coefficient of determination (R^2). Let y_t and $\hat{y}_t$ denote the observed and predicted NTDC system loads at hour t, respectively, N the number of test samples, and $\bar{y}$ the mean test load:

$$\bar{y} = \frac{1}{N}\sum_{t=1}^{N} y_t. \tag{29}$$

$$\text{NRMSE } (\%) = \frac{\sqrt{\frac{1}{N}\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}}{\bar{y}} \times 100. \tag{30}$$

$$\text{NMAE } (\%) = \frac{\frac{1}{N}\sum_{t=1}^{N}|y_t - \hat{y}_t|}{\bar{y}} \times 100. \tag{31}$$

$$\text{MAPE } (\%) = \frac{100}{N}\sum_{t=1}^{N}\left|\frac{y_t - \hat{y}_t}{y_t}\right|. \tag{32}$$

$$\text{WAPE } (\%) = 100\frac{\sum_{t=1}^{N}|y_t - \hat{y}_t|}{\sum_{t=1}^{N}|y_t|}. \tag{33}$$

$$\text{SMAPE } (\%) = \frac{100}{N}\sum_{t=1}^{N}\frac{2|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t|}. \tag{34}$$

Table 4
Hyperparameter search space.

Model family	Hyperparameter	Searched values
LSTM-based models	Hidden units	{32, 64, 128, 256}
	Number of layers	{1, 2, 3}
	Dropout	{0.0, 0.1, 0.2, 0.3}
TCN-based models (including STR-TCN and RFR-TCN)	Hidden channels	{32, 64, 128, 256}
	Number of layers	{2, 3, 4, 5}
	Kernel size	{2, 3, 5, 7}
	Dropout	{0.0, 0.1, 0.2, 0.3}
ConvLSTM/CNN-BiLSTM-type models	Convolution filters	{16, 32, 64, 128}
	Recurrent hidden size	{32, 64, 128}
	Number of layers	{1, 2, 3}
	Kernel size	{2, 3, 5}
	Dropout	{0.0, 0.1, 0.2, 0.3}
Transformer-based models	Model dimension	{64, 128, 256}
	Number of heads	{2, 4, 8}
	Number of layers	{1, 2, 3, 4}
	Feed-forward dimension	{128, 256, 512}
	Dropout	{0.0, 0.1, 0.2}
TimeXer	Patch length	{4, 6, 8, 12, 16, 24}
	Model dimension	{32, 64, 128}
	Number of heads	{2, 4, 8}
	Number of layers	{1, 2, 3}
	Dropout	{0.0, 0.1, 0.2}
S-Mamba	Model dimension	{32, 64, 128}
	State dimension	{8, 16, 32}
	Number of layers	{1, 2, 3}
	Dropout	{0.0, 0.1, 0.2}
Decomposition-based hybrid models	Number of modes/components	{3, 4, 5, 6}
	Hidden units	{32, 64, 128}
	Number of layers	{1, 2, 3}
	Dropout	{0.0, 0.1, 0.2, 0.3}

$$R^2 = 1 - \frac{\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{N}(y_t - \bar{y})^2}. \tag{35}$$

NRMSE and NMAE quantify normalized error magnitude, with NRMSE emphasizing larger deviations, whereas MAPE, WAPE, and SMAPE characterize relative forecasting error under different normalizations. R^2 measures the fraction of test-set load variability explained by the predictions. Lower error values and higher R^2 indicate stronger forecasting performance.

3.5. Ablation study

This ablation isolates the contribution of the main architectural stages of HST-RefNet over the first 6 forecast horizons. In particular, this study evaluates the standalone first-stage predictor, the standalone deep random-feature baseline, two intermediate variants that partially assemble the proposed framework, and the full HST-RefNet. Here, STR-TCN denotes the first-stage Spatio-Temporal Representation TCN used in the proposed architecture, STR-TCN + RFR-TCN denotes the hierarchical residual-refinement variant without the final calibrator, and STR-TCN + edRFC denotes the variant in which the first-stage predictor is followed directly by the embedded deep Random Feature Calibrator without the residual-refinement stage. This design separates the effects of the primary spatio-temporal representation, the residual-refinement stage, and the final closed-form calibration stage.

Table 5 summarizes the ablation results. Relative to STR-TCN, adding the residual-refinement stage reduces NRMSE from 8.81% to 6.23%, reduces MAPE from 7.56% to 6.67%, reduces SMAPE from 7.45% to 6.94%, and improves $R2$ from 0.72 to 0.84. Although NMAE and WAPE increase slightly in this variant, the clear improvement in NRMSE and $R2$ indicates that the residual-refinement stage is particularly effective in correcting larger and more difficult errors. When STR-TCN is followed directly by the edRFC module, the model achieves broader gains across all reported metrics, reducing NRMSE to 5.84%, NMAE and WAPE to 6.15%, MAPE to 5.91%, and improving $R2$ to 0.85. The full HST-RefNet delivers the strongest overall performance, achieving 5.00% NRMSE, 3.94% NMAE and WAPE, 4.02% MAPE, 4.02% SMAPE, and 0.91 $R2$. Together, these results show that the residual-refinement stage and the closed-form calibration stage contribute complementary gains, while their combination yields the most accurate and balanced forecasting performance.

Table 5
Ablation over 6-hour horizons.

Metric	STR-TCN	EDR	STR-TCN + RFR-TCN	STR-TCN + edRFC	HST-RefNet
NRMSE (%) ↓	8.81	9.36	6.23	5.84	**5.00**
NMAE (%) ↓	7.21	7.47	7.32	6.15	**3.94**
MAPE (%) ↓	7.56	7.89	6.67	5.91	**4.02**
WAPE (%) ↓	7.21	7.47	7.32	6.15	**3.94**
SMAPE (%) ↓	7.45	7.65	6.94	6.73	**4.02**
R2 ↑	0.72	0.68	0.84	0.85	**0.91**

Table 6
Average performance over first 6 horizons.

Metric	LM							LO							HST-RefNet
	CLS [48]	CRS [50]	ICB [49]	LSTM [51]	SMB [46]	TCN [52]	TMX [45]	ESE [29]	ITF [22]	LSTM [51]	SMB [46]	TCN [52]	TMX [45]	VBL [47]	
NRMSE (%) ↓	5.38	8.34	6.61	12.43	5.62	8.81	5.92	6.70	12.44	6.97	5.62	6.33	6.13	13.83	**5.00**
NMAE (%) ↓	4.19	6.92	5.29	10.05	4.45	7.21	4.62	5.34	9.90	5.56	4.37	5.05	4.88	11.69	**3.94**
MAPE (%) ↓	4.29	7.34	5.56	10.15	4.48	7.56	4.66	5.53	10.77	5.69	4.52	5.28	5.10	12.50	**4.02**
WAPE (%) ↓	4.19	6.92	5.29	10.05	4.45	7.21	4.62	5.34	9.90	5.56	4.37	5.05	4.88	11.69	**3.94**
SMAPE (%) ↓	4.28	7.18	5.48	10.42	4.52	7.45	4.71	5.52	10.44	5.68	4.46	5.23	5.04	12.15	**4.02**
R2 ↑	0.89	0.74	0.84	0.44	0.88	0.72	0.87	0.83	0.39	0.82	0.88	0.85	0.86	0.25	**0.91**

3.6. Comparison with state-of-the-art methods

This study compares HST-RefNet with recently proposed and widely used deep learning baselines for time-series prediction. All models are optimized as described in the experimental setup using TPE with 100 trials, and the selected hyperparameters are then used to train each model before reporting test-set performance. All results in Tables 5 and 6 were obtained by executing each method on this benchmark under the same chronological train/validation/test protocol and TPE search budget as HST-RefNet. To quantify the effect of meteorological inputs, standard predictors including LSTM, TCN, TimeXer (TMX) [45], and S-Mamba (SMB) [46] are trained in two settings. LO uses historical load inputs only and LM uses historical load together with the gridded meteorological tensor, except for TMX and SMB, whose LM setting instead supplies the three retained country-averaged meteorological covariates as exogenous inputs alongside the load series, consistent with how these architectures are intended to consume a modest set of distinct variates rather than an undifferentiated grid. In the load-only setting, VMD-BiLSTM-LO (VBL-LO) [47], EMD-SE-LO (ESE-LO) [29], and ITF-LO [22] are trained with load inputs only. Spatio-temporal baselines that can exploit structured meteorological fields, including ConvLSTM-LM (CLS-LM) [48], IDBO-CNN-BiLSTM (ICB) [49], and CoReSTL-LM (CRS-LM) [50], are trained with LM inputs to match their intended operating regime. This comparison protocol is designed to distinguish generic sequence models from architectures that are explicitly intended for structured spatio-temporal inputs. Accordingly, LSTM, TCN, TMX, and SMB are evaluated in both load-only and load-plus-meteorology settings to test whether direct meteorological augmentation is sufficient for these standard architectures, whereas CLS-LM and CRS-LM are included as dedicated spatio-temporal baselines designed for high-dimensional structured forecasting. Therefore, the reported gains of HST-RefNet are not inferred only from generic concatenation-based baselines, but also from comparisons against models whose architectures are specifically suited to high-dimensional spatio-temporal inputs.

Table 6 summarizes average accuracy over the 6 h horizons. HST-RefNet achieves the best overall performance with the lowest NRMSE, NMAE, MAPE, WAPE, and SMAPE and the highest R2 among the compared models. CLS-LM emerges as the strongest baseline because it directly models spatio-temporal dependence through convolutional gating and recurrent state updates, which enables effective aggregation of high-dimensional meteorological grids. Its average NRMSE of 5.38% and R2 of 0.89 exceed other LM-capable baselines such as ICB and CRS-LM, which indicates that explicit spatio-temporal structure matters in this national setting. Among the standard predictors evaluated in both settings, SMB and TMX achieve the strongest results: SMB-LO and SMB-LM both reach an NRMSE of 5.62% (R2 of 0.88), and TMX-LM reaches 5.92% and 0.87. These results remain behind HST-RefNet and CLS-LM but ahead of every other generic baseline, indicating that recent forecasting-specific architectures with a compact exogenous-variate design narrow the gap to dedicated spatio-temporal models. HST-RefNet improves upon this best competitor by reducing NRMSE to 5.00% which is a 7.09% reduction relative to CLS-LM and by increasing R2 to 0.91. These gains align with the challenges stated in the introduction. The hierarchical residual refinement targets hard operating regimes that single-stage models underfit, while the TCN-based representation avoids the state-compression limitations that can weaken recurrent models under long and heterogeneous national drivers. The embedded deep random-feature calibrator further corrects residual nonlinearities with a stable closed-form adjustment, which strengthens short-horizon accuracy without adding heavy inference cost.

The LO versus LM pairs show a consistent pattern that also supports this interpretation. Adding meteorological inputs degrades LSTM and TCN performance. NRMSE rises from 6.97% to 12.43% for LSTM and from 6.33% to 8.81% for TCN when shifting from LO to LM. This result suggests that naive concatenation of large meteorological tensors can inject noise or dilute salient temporal structure when the architecture lacks an explicit mechanism to extract and compress spatial signals. TMX and SMB do not show this degradation: TMX-LM improves over TMX-LO, and SMB-LM and SMB-LO are effectively tied, consistent with

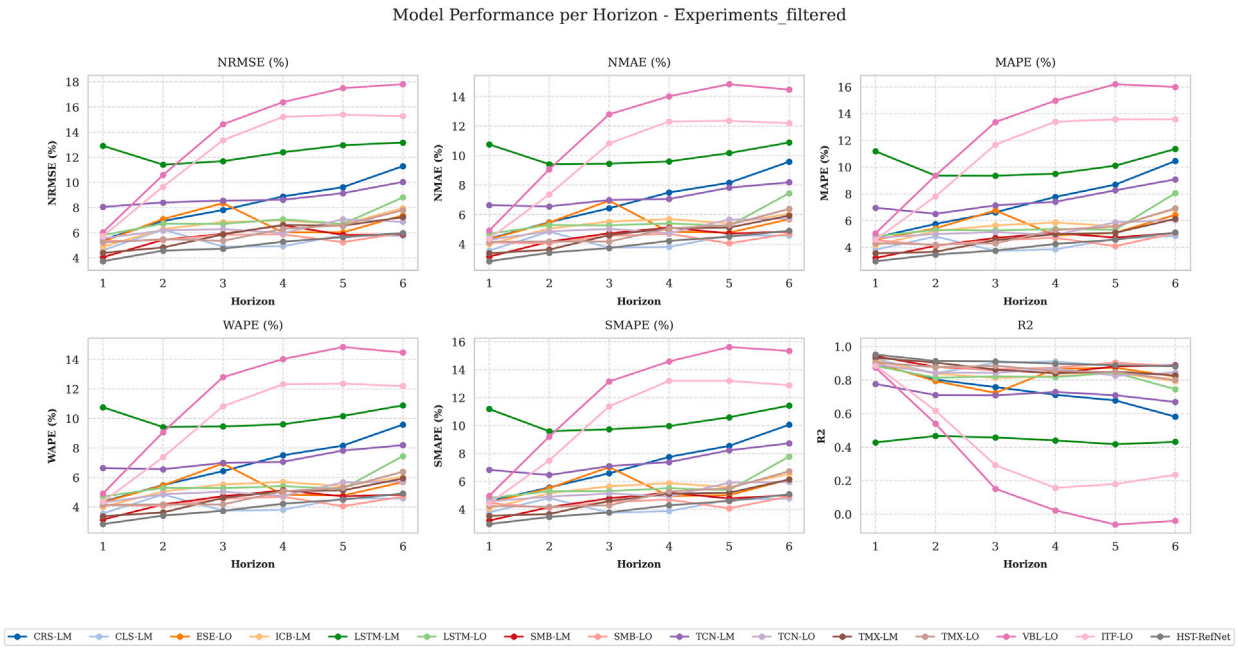


Fig. 4. Hour-by-hour comparison over the first 6 horizons for all models.

Table 7
Wilcoxon signed-rank test for HST-RefNet vs. others.

Metric (p)	LM							LO						
	CLS [48]	CRS [50]	ICB [49]	LSTM [51]	SMB [46]	TCN [52]	TMX [45]	ESE [29]	ITF [22]	LSTM [51]	SMB [46]	TCN [52]	TMX [45]	VBL [47]
NRMSE	2.46e−10	8.79e−60	1.54e−41	2.92e−63	1.83e−24	3.07e−62	1.57e−29	5.25e−55	3.44e−62	3.24e−57	1.80e−17	4.59e−49	5.08e−39	2.30e−63
NMAE	2.07e−06	7.10e−58	8.18e−40	4.44e−63	1.13e−21	1.45e−61	7.19e−24	4.00e−52	8.81e−62	6.88e−55	3.09e−13	7.91e−47	3.08e−37	1.70e−63
MAPE	3.09e−06	2.08e−59	1.63e−41	3.83e−63	2.86e−17	4.51e−62	4.78e−20	1.54e−52	5.50e−62	1.02e−54	2.29e−15	4.95e−49	4.98e−40	1.65e−63
WAPE	2.07e−06	7.10e−58	8.18e−40	4.44e−63	1.13e−21	1.45e−61	7.19e−24	4.00e−52	8.81e−62	6.88e−55	3.09e−13	7.91e−47	3.08e−37	1.70e−63
SMAPE	5.65e−06	1.80e−58	3.02e−40	4.04e−63	6.61e−19	9.64e−62	1.20e−21	4.94e−52	6.83e−62	2.71e−54	7.05e−13	6.57e−48	1.34e−38	1.81e−63
R2	2.51e−09	5.32e−57	5.36e−36	1.13e−62	6.39e−22	8.27e−61	5.73e−20	3.58e−43	6.55e−61	2.51e−51	5.57e−11	5.52e−37	3.00e−30	2.21e−62

their compact exogenous-variate design noted above. In contrast, CLS-LM remains competitive because its convolutional recurrence learns structured spatio-temporal features. HST-RefNet advances this capability by adding staged residual learning and closed-form calibration, which yields the most consistent error reduction across all reported metrics. Fig. 4 shows the hour-by-hour comparison and indicates that HST-RefNet maintains the best performance across all 6 horizons, which supports its suitability for national-scale STLF under country-wide meteorological forcing.

3.7. Statistical significance

This study validates whether the gains of HST-RefNet over competing methods are statistically consistent using the Wilcoxon signed-rank test. We apply the test to paired batch-wise metric distributions computed from the test set for NRMSE, NMAE, MAPE, WAPE, SMAPE, and $R2$. Let y_t denote the true system load and $\hat{y}_t$ denote the predicted load at time t. We form batches from the test samples, compute each metric per batch by aggregating the corresponding y_t and $\hat{y}_t$ values across the first 6 horizons, and then compare HST-RefNet against each baseline on these paired batch values. To ensure valid pairing across models, we align test samples by matching the corresponding ground-truth sequences. We report Holm-adjusted p-values to control for multiple comparisons.

Table 7 shows that all adjusted p-values remain well below 0.05 across metrics, including against the strongest baseline CLS-LM. These results support that HST-RefNet achieves statistically significant improvements over all compared methods (see Fig. 5).

3.8. Limitations and practical deployment implications

Despite the strong empirical performance of HST-RefNet, several limitations should be noted. First, the current input design does not include explicit calendar or event covariates such as hour-of-day, weekday–weekend, holiday, or festival indicators. In the present short-horizon setting, a substantial part of the intraday and recent day-type structure is captured implicitly by the 48-hour historical load window, while seasonal regime information is represented jointly by the recent load trajectory and the nationwide meteorological fields. However, rare behavioral shifts associated with holidays, festivals, or abrupt social disruptions may still be underrepresented, which can limit performance in such regimes. Second, the evaluation is conducted on a single national-scale

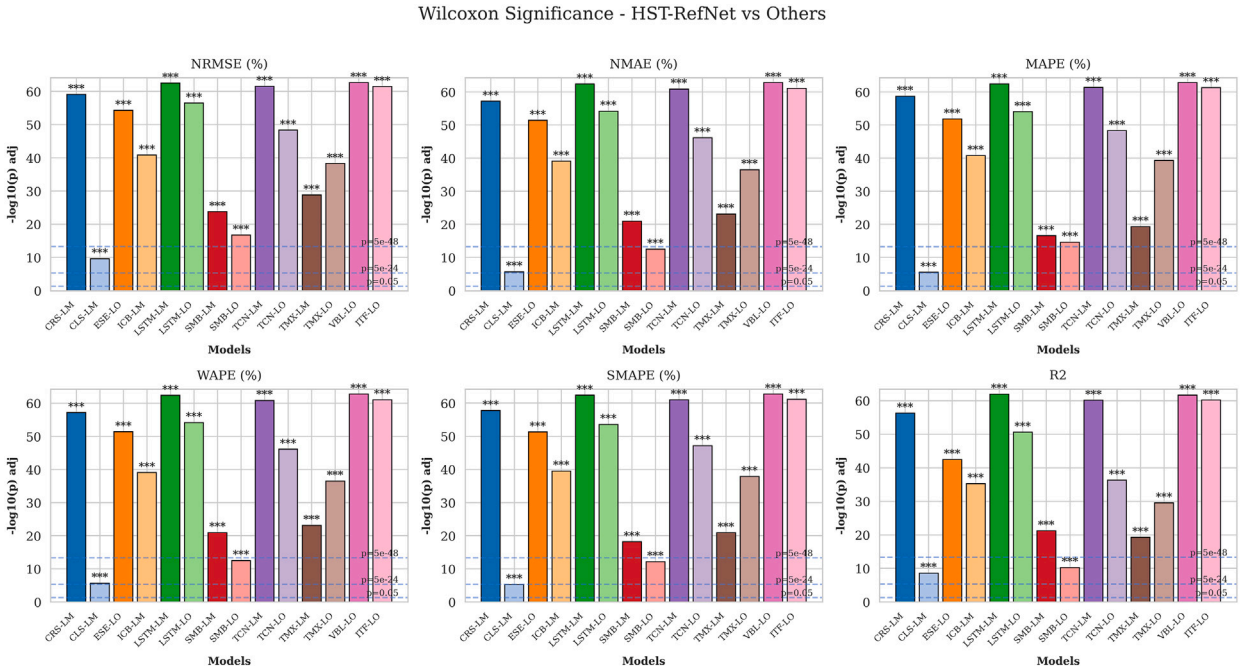


Fig. 5. Batch-wise Wilcoxon significance for HST-RefNet versus all baselines across six metrics with Holm-adjusted p-values.

dataset from Pakistan using a fixed meteorological grid resolution, so the generality of the reported gains across other countries, market structures, and spatial resolutions remains to be verified. Third, the study uses aligned NASA POWER meteorological data as the weather input source, whereas real-world deployment may depend on operational forecast products whose uncertainty can propagate into load forecasts.

From a practical deployment perspective, the proposed framework remains attractive because it uses routinely available inputs, namely recent system load and gridded meteorological data, and it avoids decomposition-heavy preprocessing. The STR-TCN and RFR-TCN stages support efficient parallel inference, and the embedded edRFC layer adds a lightweight closed-form calibration step with low additional computational cost. These properties make HST-RefNet suitable for short-horizon national or large-area dispatch support, provided that a stable weather data pipeline and periodic retraining or recalibration are maintained. In practical use, explicit calendar and event indicators can also be incorporated as an extension to further improve robustness during special-demand periods.

4. Conclusion

This study proposed the Hierarchical Spatio-Temporal Refinement Network (HST-RefNet) for national-scale short-term load forecasting by integrating Pakistan-wide gridded meteorological fields with historical NTDC demand and refining predictions of y_t into $\hat{y}_t$ through staged temporal convolution and lightweight closed-form calibration. The results showed that the hierarchical residual design and embedded deep random-feature correction jointly improved accuracy across NRMSE, NMAE, MAPE, WAPE, SMAPE, and $R2$ when compared with strong spatio-temporal and load-only baselines, and the batch-wise Wilcoxon signed-rank test with Holm adjustment confirmed that the gains were statistically consistent. However, the evaluation was limited to a single Pakistan-based dataset and a fixed grid resolution, so the generality of the proposed architecture across other countries, market structures, climate regimes, and spatial resolutions remains to be established, particularly because similarly large paired load-meteorology datasets remain scarce. Future work will extend validation to multi-country and multi-resolution settings, including probabilistic and transfer-learning variants, to test robustness under broader operational conditions.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Fang Liu reports financial support was provided by National Natural Science Foundation of China. Fang Liu reports financial support was provided by Key Projects of Hunan Provincial Department of Education. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The meteorological data supporting this study are publicly available from NASA's POWER project at https://power.larc.nasa.gov. The Pakistan national load data from NTDC are available from the corresponding author upon reasonable request.

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